Deep Neural Networks for Learning Intent from sEMG Signals to Support Hardware Devices for Post-Stroke Neurorehabilitation

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Abstract

Finger-specific motor intent is a clinically meaningful target for post-stroke neurorehabilitation because recovery of fine hand control often depends on detecting weak, variable, and patient-specific muscle activity. This project studies five-finger intent decoding from high-density surface electromyography (sEMG) using PhysioMio, a bilateral longitudinal dataset collected from stroke patients [8]. We formulate the task as multilabel prediction from impaired-arm sEMG segments and evaluate recurrent, convolutional, and graph-based models under a shared signal-processing pipeline. Raw recordings are label-aligned, band-pass filtered from 20–450 Hz, denoised with Symlet-4 wavelets, segmented into 200 ms windows with 50% overlap, and represented by twelve time- and frequency-domain descriptors per channel-window pair. The resulting tensors support direct comparison among LSTM, CNN, and GNN decoders as well as subsequent optimization of compact CNN students and ResNet teachers for Raspberry Pi 5 hardware deployment [15, 16]. In the initial three-model comparison, the LSTM obtains the highest exact-match accuracy (0.545), while the GNN obtains the strongest macro F1 (0.706) and macro AUPRC (0.776), indicating distinct temporal and structural advantages. Architecture search over the CNN family then moves the empirical frontier: the Optuna-tuned CNN-Large student reaches 0.593 exact-match accuracy, 0.798 finger accuracy, 0.714 macro F1, 0.881 macro AUROC, and 0.812 macro AUPRC, slightly exceeding the best ResNet teacher on most aggregate metrics. Healthy-to-impaired transfer learning further improves the CNN branch but produces mixed results for the LSTM, suggesting that transfer gains are architecture-dependent. Finally, we describe the implemented distillation and latency-measurement path that connects the current models to embedded inference. Together, these results define a software-centered research basis for finger-intent decoding in post-stroke rehabilitation systems.

1 Introduction

Stroke frequently disrupts the descending motor pathways needed for coordinated hand opening, grasping, and individuated finger control. For patients with residual forearm muscle activity, surface electromyography (sEMG) offers a non-invasive sensing modality through which intended movement can be estimated even when visible motion is weak or incomplete [13, 20, 4]. Reliable finger-intent decoding is therefore a central software problem for rehabilitation devices that aim to provide timely assistance, feedback, or task-specific practice.

The decoding problem is difficult because post-stroke sEMG differs substantially from healthy-limb gesture-recognition benchmarks. Paretic recordings are affected by abnormal co-contraction, altered recruitment patterns, fatigue, sensor placement, and impairment severity. In addition, the intended output for rehabilitation is often not a single gesture class but a multi-finger activation pattern suitable for controlling or cueing assistive hardware. A useful model must therefore be evaluated together with the signal-processing and labeling decisions that define the task.

This project studies that problem through a software pipeline developed for a post-stroke rehabilitation system [15]. The broader goal is to serve compact machine-learning models on Raspberry Pi 5 hardware and connect their predictions to a finger-assist device. The present paper focuses on the scientific and computational layer: how impaired-arm sEMG is transformed into five-finger labels, how several model families behave under a common representation, and how transfer learning and compact CNN optimization change the resulting performance profile.

Our study is organized around two questions. First, when LSTM, CNN, and GNN decoders are trained on the same PhysioMio-derived feature representation, which inductive biases are most useful for multilabel finger-intent prediction? Second, after the baseline comparison, how much can the system be improved through compact CNN architecture search, teacher models, and healthy-to-impaired transfer learning? This organization reflects the intended research progression: establish the behavior of individual model families first, then examine optimization strategies that support practical inference.

The contributions are fourfold. We define a reproducible sEMG-to-finger-intent pipeline for PhysioMio recordings, including label alignment, filtering, denoising, windowed feature extraction, patient-level splits, and shared tensor adaptation. We compare LSTM, CNN, and GNN predictors under this common formulation and report both aggregate and finger-wise multilabel metrics. We evaluate an optimized CNN student family against internal ResNet teachers and quantify the performance-size trade-off relevant to embedded deployment. Finally, we analyze healthy-to-impaired transfer learning and the implemented distillation path as mechanisms for improving accuracy and inference efficiency in future rehabilitation systems.

2 Related Work

sEMG has long been studied as a non-invasive control channel for rehabilitation, prosthetics, and human-machine interaction, and stroke-specific reviews show that sEMG-driven interventions can support meaningful upper-limb rehabilitation when signal interpretation is reliable enough for assistive use [13, 20]. Broader reviews of wearable sensing for stroke recovery make a related point: clinical value depends on the full stack, including acquisition quality, body placement, labeling protocol, feature representation, and the way machine learning is coupled to assessment or assistive control [4]. For this reason, decoding architecture should not be discussed independently of data construction and deployment context.

Dataset choice is especially important in this area. Ninapro remains the dominant public benchmark family for hand-gesture decoding from sEMG, with ten sub-datasets spanning different electrode layouts, movement vocabularies, and sensor modalities [3, 14]. The widely used DB5 split, for example, records 52 movements from 10 healthy participants with two Myo armbands, while DB1 and DB2 cover larger healthy cohorts with different channel configurations and richer kinematic metadata [3, 14]. By contrast, PhysioMio was introduced in 2026 as a stroke-specific bilateral and longitudinal HD-sEMG resource with 48 stroke patients, 16 gestures, 64 electrodes, and repeated recordings across inpatient recovery [8]. That distinction matters because results on healthy-subject multiclass gesture sets are not automatically transferable to impaired-arm intent decoding in post-stroke populations.

Stroke-specific decoding studies have historically been small and heterogeneous. Lee et al. [10]

used subject-specific myoelectric pattern classification for six functional hand movements in chronic stroke survivors and reported a sharp impairment-dependent gap, with mean accuracy of 71.3% for moderately impaired participants and 37.9% for severely impaired participants. Anastasiev et al. [2] later studied post-acute stroke gesture recognition with portable eight-channel sEMG and found that affected-side SVM performance could still reach the high-80% range on reduced gesture sets, but the task remained sensitive to feature design, label count, and paretic signal quality. These studies underscore a recurring challenge for post-stroke intent decoding: strong performance is possible, but dataset scale and protocol differences make general claims difficult.

Deep neural models broaden that picture but do not remove the comparability problem. Sequence-oriented recurrent models remain natural baselines because they aggregate temporal evidence across windows, while compact 1D CNNs emphasize local temporal motifs and can be compressed aggressively for efficient inference [7]. Graph neural networks offer a complementary structural view by representing windows as related observations connected through message passing rather than as a purely linear sequence [9, 5, 18]. On healthy-dataset benchmarks such as Ninapro, deeper end-to-end networks can reach very high multiclass accuracies; for example, Sri-Iesaranusorn et al. [17] reported 93.87% accuracy on Ninapro DB5 and 91.69% on DB7 for 41-movement classification. Those results are useful context for model capacity, but they address a different population and a different task than impaired-arm multilabel finger-intent prediction.

Transfer learning has recently become one of the most relevant directions for rehabilitation-oriented sEMG. Li et al. [11] showed that transfer-based CNNs can materially improve inter-subject and inter-day robustness on high-density healthy-subject sEMG, and Mohammadiazni et al. [12] demonstrated an especially important stroke-specific result: pretraining on healthy Ninapro DB5 data raised stroke-survivor gesture-classification accuracy from 46% to 93.6% in a three-gesture setting. These findings motivate healthy-to-impaired transfer as a natural strategy for paretic-limb decoding, where affected-side training data are comparatively scarce and variable.

Deployment-oriented optimization provides the final piece of context for this manuscript. Knowledge distillation is a standard way to transfer behavior from larger teachers into smaller students [6], and Optuna is widely used for black-box hyperparameter search under practical constraints [1]. For rehabilitation wearables, these methods connect offline decoding accuracy to the practical requirements of model size, latency, and hardware-constrained inference.

3 Method

3.1 Overview

This project defines a modular decoding workflow from multichannel sEMG and gesture annotations to five-finger prediction logits. The pipeline contains four stages: data harmonization, signal processing, model inference, and evaluation or deployment analysis. Figure 1 summarizes the workflow used throughout the study.

For notation, let a pre-segmented recording be $X \in \mathbb{R}^{T\times C}$, where T is the number of raw time samples and $C = 64$ is the channel count under the PhysioMio path. After preprocessing and windowing, the project extracts a feature tensor $\mathcal{X} \in \mathbb{R}^{C\times W\times F}$, where W is the number of windows and $F = 12$ is the number of handcrafted descriptors per channel-window pair. Each sample is paired with a multilabel target vector $y \in \{0, 1\}^5$ indicating intended activation of thumb, index, middle, ring, and little finger.

Table 1: Literature context for the current study. Reported numbers reflect different populations, sensors, gesture vocabularies, and label spaces.

Source	Population / dataset	Task	Reported performance	Relevance to this paper
Lee et al. 2010	20 chronic stroke survivors; 10 forearm/hand electrodes	Six functional hand movements with subject-specific EMG classification	Mean accuracy 71.3% for moderate impairment and 37.9% for severe impairment	Early evidence that stroke severity strongly affects decoder quality
Anastasiev et al. 2022	19 post-acute stroke patients; portable eight-channel sEMG	Four- to seven-gesture multiclass recognition on affected and non-affected sides	Affected-side SVM accuracy reached 88.7% on the four-gesture setting	Shows that reduced-vocabulary stroke gesture recognition can reach high accuracy with careful feature design
Mohammadiazni et al. 2026	10 stroke patients plus healthy Ninapro DB5 pretraining data	Three-gesture classification across multiple arm postures with transfer learning	93.6% with transfer learning versus 46% without transfer	Strong stroke-specific motivation for healthy-to-impaired transfer learning
Sri-Iesaranusorn et al. 2021	Ninapro DB5/DB7, mostly healthy-subject settings	41-movement deep neural network classification	93.87% on DB5 and 91.69% on DB7	Healthy-data ceiling is high, but the task is easier than impaired-arm multilabel finger decoding
This project	PhysioMio impaired-arm split from 48 stroke patients; patient-level train/val/test split	Five-finger multilabel intent decoding from processed HD-sEMG segments	Best model: CNN-Large with 0.593 exact-match accuracy and 0.714 macro F1	Stricter label space and patient split make this a harder but more rehabilitation-specific benchmark

3.2 Data harmonization and ingestion

The implementation builds processed datasets directly from raw PhysioMio parquet files. The loader searches the raw data tree, groups files by patient, aligns contiguous stretches of the same `movement_type`, maps each label to a five-bit finger target, discards segments shorter than 200 samples, and applies the preprocessing pipeline to each usable segment. This converts gesture-level clinical recordings into segment-level multilabel examples.

The default configuration uses PhysioMio recordings sampled at 2000 Hz and selects the impaired arm. Patients are partitioned into deterministic 70/10/20 train/validation/test splits, so samples from the same patient do not appear in multiple partitions. Processed samples are padded to a dataset-wide maximum window count for batched tensor operations.

3.3 Signal preprocessing

The preprocessing module is explicitly parameterized by a frozen configuration object in the implementation. Raw sEMG first passes through a fourth-order Butterworth band-pass filter with 20–450 Hz cutoffs and zero-phase forward-backward application. This stage suppresses low-frequency motion artifacts and high-frequency electrical noise while preserving time alignment across channels.

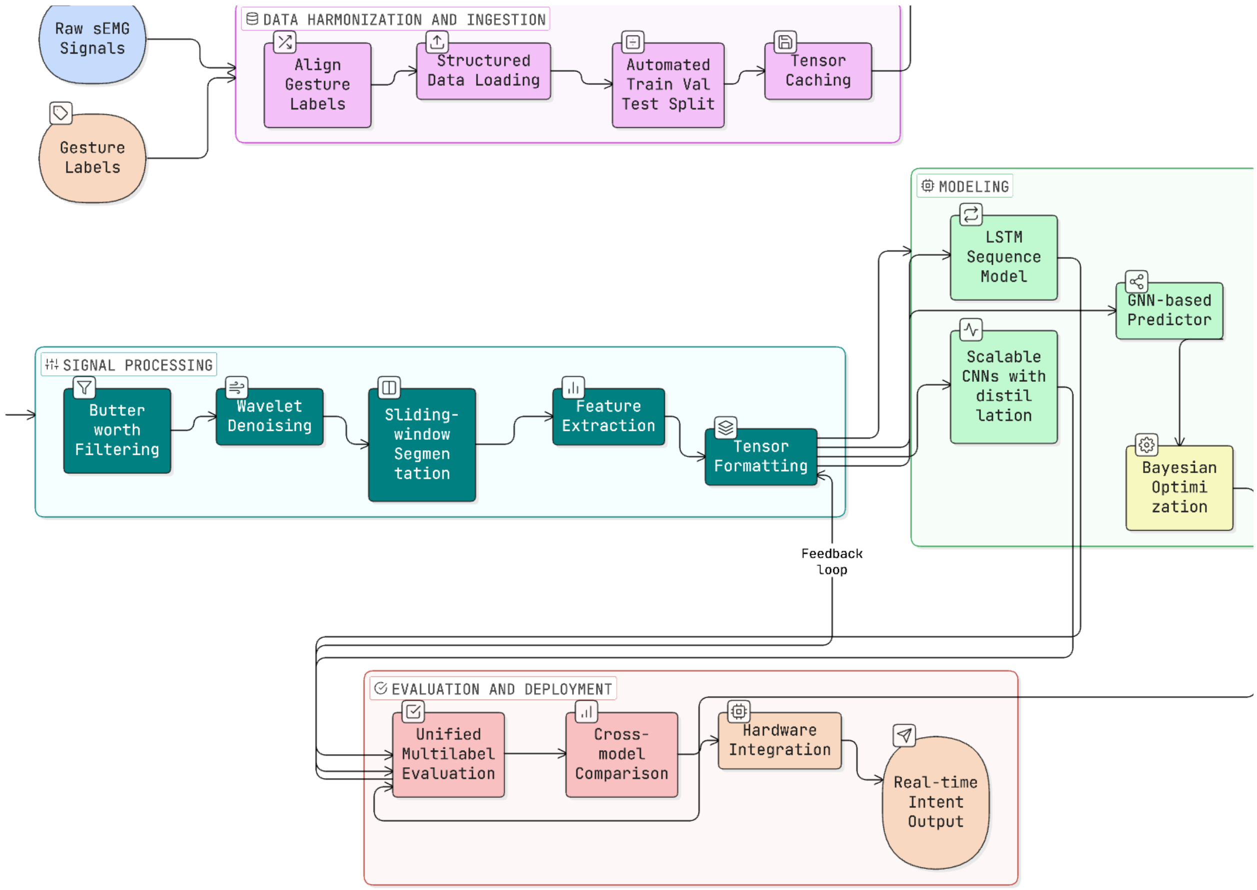

Figure 1: Overview of the sEMG finger-intent decoding pipeline. Raw sEMG and gesture annotations are aligned, segmented, featurized, mapped into shared tensors, evaluated across multiple model families, and prepared for downstream deployment analysis.

The filtered signal is then denoised with wavelet soft-thresholding. The default configuration uses a Symlet-4 basis, four decomposition levels, universal threshold selection, and a median-based noise estimate. The same preprocessing configuration is used for all model families, making downstream differences attributable to the representation and predictor rather than to separate signal-conditioning choices.

After denoising, each continuous segment is divided into 200 ms windows with 50% overlap and no zero-padding. These settings balance temporal responsiveness against the sample count required for stable summary statistics.

3.4 Window-level feature representation

Each channel-window pair is summarized by twelve handcrafted descriptors spanning time and frequency domains. The time-domain set includes root mean square, mean absolute value, integrated EMG, waveform length, variance, zero crossings, slope sign changes, and Willison amplitude. The frequency-domain set includes mean frequency, median frequency, spectral entropy, and total power, with spectral quantities estimated from Welch periodograms using `nperseg = min(256, len(x))` [19]. Degenerate spectra are zeroed explicitly to avoid unstable feature values. The resulting tensor has shape (C, W, F), where C is the number of channels, W is the number of windows, and $F = 12$ is the feature count.

3.5 Tensor formatting and shared adapter

The project standardizes cross-model comparison with one shared adapter. It transforms (C, W, F) or (N, C, W, F) tensors into $(N, W, C \times F)$ sequences, where windows act as time steps and all channel features are concatenated into one vector per step. For a single sample, the adapted sequence can be written as

$$S = [s_1, \ldots, s_W]^\top \in \mathbb{R}^{W \times d}, \qquad s_w = \mathrm{vec}(\mathcal{X}_{:,w,:}), \qquad d = CF = 768.$$

This adapter creates a common input space for recurrent, convolutional, and graph-based predictors.

For the CNN branch, the adapted tensor is subsequently permuted to a channels-first layout. The CNN stack treats the 64×12 channel-feature grid as 768 input channels for temporal `Conv1d` layers, preserving the same upstream representation while allowing convolutions over window order.

3.6 Predictor families

The LSTM branch applies two stacked recurrent layers with hidden sizes 128 and 256, followed by a 128-unit fully connected layer and dropout before five output logits. Given adapted sequence input s_t, the recurrent dynamics follow the standard LSTM update

$$\begin{aligned}
i_t &= \sigma(W_i s_t + U_i h_{t-1} + b_i), \\
f_t &= \sigma(W_f s_t + U_f h_{t-1} + b_f), \\
o_t &= \sigma(W_o s_t + U_o h_{t-1} + b_o), \\
\tilde{c}_t &= \tanh(W_c s_t + U_c h_{t-1} + b_c), \\
c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \\
h_t &= o_t \odot \tanh(c_t),
\end{aligned}$$

and the project uses the final hidden state from the second recurrent layer for classification. This branch emphasizes temporal accumulation across windows and serves as the main sequential baseline for the direct benchmark.

The GNN branch interprets each time window as a node in a graph. The implementation supports GCN, GraphSAGE, and GAT operators [9, 5, 18], and the benchmarked path constructs a dense graph over windows within each sample before graph-level pooling and readout. At a generic level, the node update can be written as

$$h_i^{(\ell+1)} = \psi_\ell\left(h_i^{(\ell)}, \operatorname*{AGG}_{j \in \mathcal{N}(i)} \phi_\ell\left(h_i^{(\ell)}, h_j^{(\ell)}\right)\right),$$

followed by graph pooling

$$g = \operatorname*{POOL}_{i=1}^{W} h_i^{(L)}$$

and a multilabel readout head. This branch injects an explicit structural prior: windows are related observations rather than only a flat sequence.

The CNN branch includes five student variants ranging from Nano to XLarge, each built around a 1×1 projection, temporal convolution blocks, adaptive pooling, and a compact fully connected head. If the channels-first representation is denoted by $\hat{S} \in \mathbb{R}^{d \times W}$, the student stack can be summarized as

$$H^{(0)} = \rho\Big(\mathrm{BN}(W_p * \hat{S} + b_p)\Big), \qquad H^{(\ell+1)} = \rho\Big(\mathrm{BN}(W_\ell * H^{(\ell)} + b_\ell)\Big),$$

where $*$ denotes 1D convolution over window order and ρ is a ReLU nonlinearity. Adaptive pooling compresses the temporal axis before a multilayer perceptron produces the five logits. The student family spans approximately 54K parameters and 53 KB INT8 for Nano to 1.62M parameters and 1.58 MB INT8 for XLarge. The teacher family uses 1D ResNet-50, ResNet-101, and ResNet-152 variants with bottleneck residual blocks.

3.7 Training, tuning, and evaluation protocol

The training utilities unify optimization and reporting across model families. The current implementation uses binary cross-entropy with logits for five-label prediction, Adam-based optimization, plateau-triggered learning-rate reduction, early stopping, checkpointing, and saved metric curves. For logits $z \in \mathbb{R}^5$, the base multilabel objective is

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{5}\sum_{k=1}^{5}\left[y_k \log \sigma(z_k) + (1 - y_k)\log\left(1 - \sigma(z_k)\right)\right].$$

Evaluation reports exact-match accuracy, finger-level averaged accuracy, macro precision, macro recall, macro F1, macro AUROC, and macro AUPRC. The metrics module also stores per-finger scores and task-level curve data, which enables both aggregate and finger-wise analysis after training.

For the CNN students, Optuna searches architecture and training choices for each model size, and the optimization score combines validation accuracy with a latency-based penalty relative to a baseline timing reference [1]. Trial-level early stopping is used during search. The system also supports teacher-ensemble distillation, per-finger threshold tuning, and latency benchmarking. For multilabel distillation, the implemented loss blends the hard target term with a temperature-scaled soft target from the teacher,

$$\mathcal{L}_{\mathrm{KD}} = \alpha\,\mathcal{L}_{\mathrm{BCE}}(z_s, y) + (1 - \alpha)T^2\,\mathrm{BCELogits}\left(\frac{z_s}{T}, \sigma\left(\frac{z_t}{T}\right)\right),$$

where z_s and z_t are student and teacher logits, T is the distillation temperature, and α controls the hard-soft balance. A separate latency benchmark reports mean, median, and 95th-percentile inference time on a chosen device.

3.8 Transfer-learning path

In addition to direct impaired-arm benchmarks, the study evaluates two-stage CNN and LSTM training. Models are first pretrained on healthy-arm recordings and then finetuned on impaired-arm recordings while preserving the same five-finger multilabel objective. This tests whether healthy-limb structure can provide a useful initialization for paretic-limb decoding.

4 Experiments

4.1 Dataset and split construction

All experiments use the PhysioMio processing path implemented for this project [8, 15]. The dataset construction step extracts contiguous movement segments from raw parquet recordings, maps gesture labels to five-bit finger targets, preprocesses each segment, and stores processed tensors as PyTorch split files.

Patients are shuffled with a fixed seed and divided into 70% train, 10% validation, and 20% test partitions. The main benchmark uses impaired-arm recordings. The transfer-learning experiments use healthy-arm recordings for pretraining and impaired-arm recordings for finetuning.

4.2 Model groups

The experiments are organized into three groups. The first group compares direct LSTM, CNN, and GNN baselines trained under the same preprocessing and feature representation. The second group evaluates optimized CNN students and ResNet teachers, emphasizing the trade-off between multilabel performance and model footprint. The third group evaluates healthy-to-impaired transfer learning and describes distillation as the compression path connecting teacher models to compact students.

The direct baselines and optimized CNN evaluations were produced at different stages of the study. We therefore report them in separate tables rather than collapsing them into a single undifferentiated leaderboard. This preserves the scientific sequence of the work: baseline comparison first, model-family refinement second.

4.3 Evaluation protocol

The task is five-label multilabel classification with a nominal decision threshold of 0.5 on the held-out test split. We report exact-match accuracy, mean finger accuracy, macro precision, macro recall, macro F1, macro AUROC, macro AUPRC, and per-finger metrics. Exact-match accuracy is strict because all five labels must be correct for a sample to count as correct; finger accuracy and macro metrics expose behavior that exact match alone can hide.

The Results section follows the experimental sequence: direct baselines, optimized CNN students and ResNet teachers, transfer learning, finger-level behavior, and literature positioning.

5 Results

5.1 Direct baseline models: LSTM, CNN, and GNN

Table 2 reports the direct comparison among LSTM, CNN, and GNN decoders on the impaired-arm test split. The three families express different assumptions about the same windowed feature sequence. The LSTM treats the input as an ordered temporal process, the CNN searches for local temporal motifs after channel-feature projection, and the GNN treats windows as nodes in a sample-level graph. Their results show complementary behavior: the LSTM leads exact-match accuracy and macro AUROC, while the GNN leads macro F1 and macro AUPRC.

Table 2: Held-out test performance for the three direct model families. Bold numbers mark the best score in each column within the baseline comparison.

Model	Exact-match	Finger Acc.	Macro F1	Macro AUROC	Macro AUPRC
LSTM	**0.545**	**0.784**	0.705	**0.858**	0.754
CNN legacy	0.442	0.694	0.676	0.767	0.764
GNN	0.448	0.683	**0.706**	0.787	**0.776**

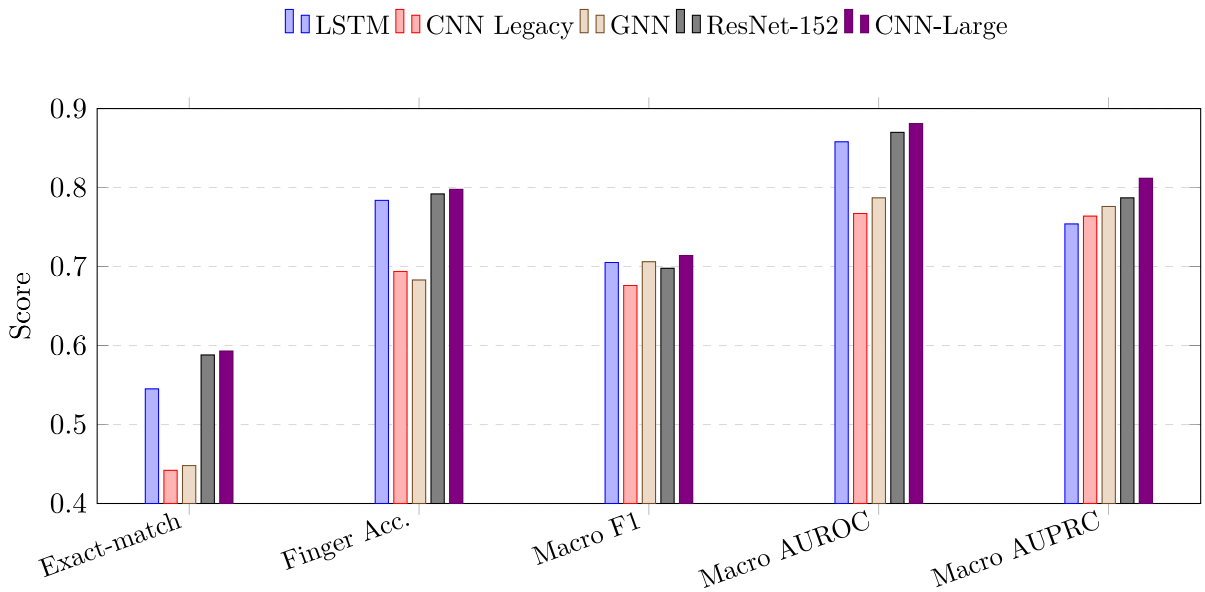

Figure 2: Aggregate-metric comparison across the three direct baselines plus the strongest internal teacher and student references from the newer CNN branch.

The baseline comparison suggests that temporal recurrence and graph aggregation capture different aspects of paretic sEMG. The LSTM's stronger exact-match accuracy indicates better all-label consistency, whereas the GNN's macro F1 and AUPRC suggest stronger class-balanced discrimination. The original CNN is weaker in this stage, motivating a more systematic convolutional architecture search.

5.2 Performance-improvement path: optimized CNN students and ResNet teachers

Table 3 evaluates the optimized CNN student family together with the internal ResNet teacher models. Relative to the direct CNN baseline, the optimized students improve every major aggregate metric. The strongest student, CNN-Large, reaches 0.593 exact-match accuracy, 0.798 finger accuracy, 0.714 macro F1, 0.881 macro AUROC, and 0.812 macro AUPRC.

The optimized CNN results reveal a useful performance-size structure. CNN-Large improves exact-match accuracy from 0.442 to 0.593 relative to the direct CNN baseline, while macro AUROC rises from 0.767 to 0.881 and macro AUPRC from 0.764 to 0.812. Nano and Micro remain close to the larger students despite substantially smaller footprints, suggesting that the handcrafted feature representation is compatible with compact temporal convolution. ResNet-152 is the strongest

Table 3: CNN-family evaluation. Student parameter counts and INT8 sizes describe the deployment-oriented model scale; ResNet rows provide high-capacity teacher references. Bold numbers mark the best score in each column.

Variant	Type	Params	INT8 Size	Exact-match	Finger Acc.	Macro F1	Macro AUROC	Macro AUPRC
ResNet-50	Teacher	–	–	0.570	0.788	0.656	0.840	0.772
ResNet-101	Teacher	–	–	0.569	0.795	0.681	0.852	0.767
ResNet-152	Teacher	–	–	0.588	0.792	0.698	0.870	0.787
Nano	Student	54K	53 KB	0.580	0.794	0.698	0.855	0.758
Micro	Student	158K	154 KB	0.578	0.794	0.697	0.871	0.793
Base	Student	390K	381 KB	0.575	**0.798**	0.707	0.870	0.775
Large	Student	803K	784 KB	**0.593**	**0.798**	**0.714**	**0.881**	**0.812**
XLarge	Student	1.62M	1.58 MB	0.551	0.794	0.690	0.830	0.726

Table 4: Healthy-to-impaired transfer-learning summaries. The CNN branch improves consistently after finetuning; the LSTM branch improves on exact-match and finger accuracy but declines on ranking-sensitive metrics.

Model / Stage	Exact-match	Finger Acc.	Macro F1	Macro AUROC	Macro AUPRC
CNN pre.	0.465	0.760	0.664	0.833	0.699
CNN fine.	**0.585**	**0.800**	**0.680**	**0.863**	**0.767**
LSTM pre.	0.523	0.752	**0.679**	**0.858**	**0.773**
LSTM fine.	**0.574**	**0.777**	0.660	0.830	0.736

teacher with 0.588 exact-match accuracy and 0.698 macro F1; CNN-Large slightly exceeds it on exact match, macro F1, macro AUROC, and macro AUPRC, indicating that the tuned student family is not merely a compressed approximation of the teacher line but a competitive model class in its own right.

5.3 Improvement paths for metrics and speed

Table 4 summarizes healthy-to-impaired transfer learning. For the CNN branch, finetuning improves exact-match accuracy from 0.465 to 0.585, finger accuracy from 0.760 to 0.800, macro AUROC from 0.833 to 0.863, and macro AUPRC from 0.699 to 0.767. The LSTM branch behaves differently: exact-match and finger accuracy improve after finetuning, but macro F1, AUROC, and AUPRC decrease. Transfer learning therefore appears beneficial for the convolutional branch while exposing architecture-dependent sensitivity in the recurrent branch.

The teacher-student design also provides a path toward faster inference. The current distillation objective combines hard multilabel supervision with temperature-scaled teacher probabilities, allowing compact students to learn from ResNet teachers while retaining the same five-finger output space. Because the present study reports the optimized student and teacher evaluations separately, distillation is treated as the next compression experiment rather than as the source of the student results in Table 3.

5.4 Finger-level behavior and external positioning

Finger-wise scores show that model ranking varies across outputs. Table 5 compares the LSTM direct baseline, the GNN direct baseline, and the best optimized CNN student. The GNN is

Table 5: Per-finger F1 score by model family. The optimized CNN-Large replaces the legacy direct CNN row for the finger-level comparison. Bold numbers mark the strongest model for each finger.

Finger	LSTM	GNN	CNN-Large
Thumb	0.716	**0.776**	0.729
Index	0.723	0.696	**0.750**
Middle	**0.712**	0.689	0.698
Ring	**0.696**	0.688	0.695
Little	0.680	0.678	**0.696**

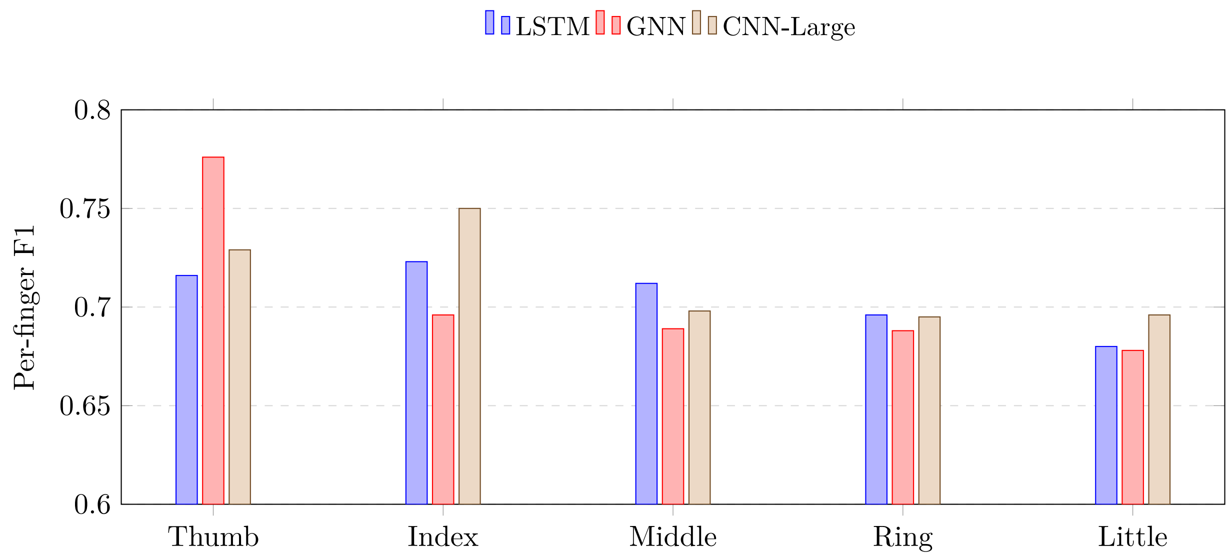

Figure 3: Per-finger F1 comparison derived from the committed metric files. Aggregate CNN gains coexist with output-specific strengths for the LSTM and GNN baselines.

strongest on thumb, CNN-Large leads on index and little finger, and the LSTM remains slightly strongest on middle and ring finger. These differences are clinically relevant because a rehabilitation controller may value different fingers differently depending on the movement being assisted.

Table 1 places the results beside prior stroke and sEMG studies. The comparison is not a leaderboard because the tasks differ substantially across datasets and populations. The present task is stricter than many gesture-classification settings: it uses impaired-arm PhysioMio recordings, patient-level splits, and a five-label exact-match metric that requires all finger activations to be correct simultaneously. Within that setting, the optimized CNN branch provides the strongest current aggregate result while the baseline and per-finger analyses show why multiple inductive biases remain scientifically informative.

6 Discussion

The results support three main observations about post-stroke finger-intent decoding from HD-sEMG. First, no single inductive bias is uniformly superior in the direct baseline setting. The LSTM's advantage on exact-match accuracy suggests that recurrent state can help produce coherent multi-finger predictions across a segment, while the GNN's macro F1 and AUPRC indicate strong class-balanced discrimination when windows are modeled as related observations. This is consistent with the structure of the task: paretic sEMG contains both temporal continuity and non-local relationships among windows.

Second, the optimized CNN results show that compact temporal convolution can be highly effective once the architecture is tuned for the feature representation. The gap between the direct CNN baseline and CNN-Large is large enough to change the empirical interpretation of the model family. Rather than treating CNNs as a weak baseline, the tuned student sweep suggests that temporal convolution is a strong candidate for embedded finger-intent decoding, especially because Nano and Micro preserve competitive performance at small estimated INT8 footprints.

Third, transfer learning appears useful but architecture-dependent. The CNN branch benefits consistently from healthy-to-impaired staging, improving exact-match, finger accuracy, macro AUROC, and macro AUPRC. The LSTM branch improves on thresholded accuracy but loses ranking-sensitive performance after finetuning. This pattern aligns with recent stroke-specific transfer-learning work that uses healthy sEMG to compensate for limited paretic data [12], while also showing that transfer must be evaluated separately for each model family.

The deployment implications are encouraging. The optimized students are small enough to motivate Raspberry Pi 5 inference, and the implemented distillation objective creates a principled mechanism for transferring teacher behavior into compact models. Threshold tuning and latency benchmarking complete the basic software ingredients required for future closed-loop experiments. The next scientific step is to measure this full chain under device-level timing constraints and to determine whether distilled students can preserve the CNN-Large performance profile at smaller footprint and lower latency.

The study has several limitations. It uses one primary dataset and one main preprocessing configuration, so additional ablations are needed to isolate the effects of filtering, denoising, feature selection, graph construction, and threshold choice. Literature comparisons remain contextual because published sEMG studies vary widely in subject population, electrode layout, gesture vocabulary, and label formulation. Finally, the present results evaluate decoding performance, not therapeutic outcome; human-subject studies with hardware-in-the-loop feedback will be required to connect predicted intent to rehabilitation benefit.

7 Conclusion

This paper studied five-finger intent decoding from post-stroke HD-sEMG using a shared signal-processing and tensor-adaptation pipeline. Direct LSTM, CNN, and GNN baselines showed complementary strengths, with the LSTM leading exact-match accuracy and the GNN leading macro F1 and AUPRC. Subsequent CNN architecture search substantially improved the convolutional branch: CNN-Large reached the strongest aggregate performance while remaining within an embedded-oriented model scale, and smaller students preserved competitive accuracy at much lower estimated INT8 size.

Healthy-to-impaired transfer learning further improved the CNN branch but produced mixed LSTM behavior, highlighting the architecture-specific nature of transfer in paretic sEMG decoding. The resulting system provides a concrete foundation for future work on distilled students, device-level latency measurement, and hardware-in-the-loop rehabilitation studies. More broadly, the study shows that post-stroke finger-intent decoding benefits from treating signal processing, model family, transfer strategy, and deployment constraints as a single experimental design problem.

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